

Hyperparameters and Validation

K Arnold

```
data(ames, package = "modeldata")
ames_all <- ames %>%
  filter(Gr_Liv_Area < 4000, Sale_Condition == "Normal") %>%
  mutate(across(where(is.integer), as.double)) %>%
  mutate(Sale_Price = Sale_Price / 1000)
rm(ames)
```

```
set.seed(10) # Seed the random number generator
ames_split <- initial_split(ames_all, prop = 2 / 3)
ames_train <- training(ames_split)
ames_test <- testing(ames_split)
```

```
model1 <-
  decision_tree(mode = "regression", tree_depth = 2) %>%
  fit(Sale_Price ~ Latitude + Longitude, data = ames_train)
```

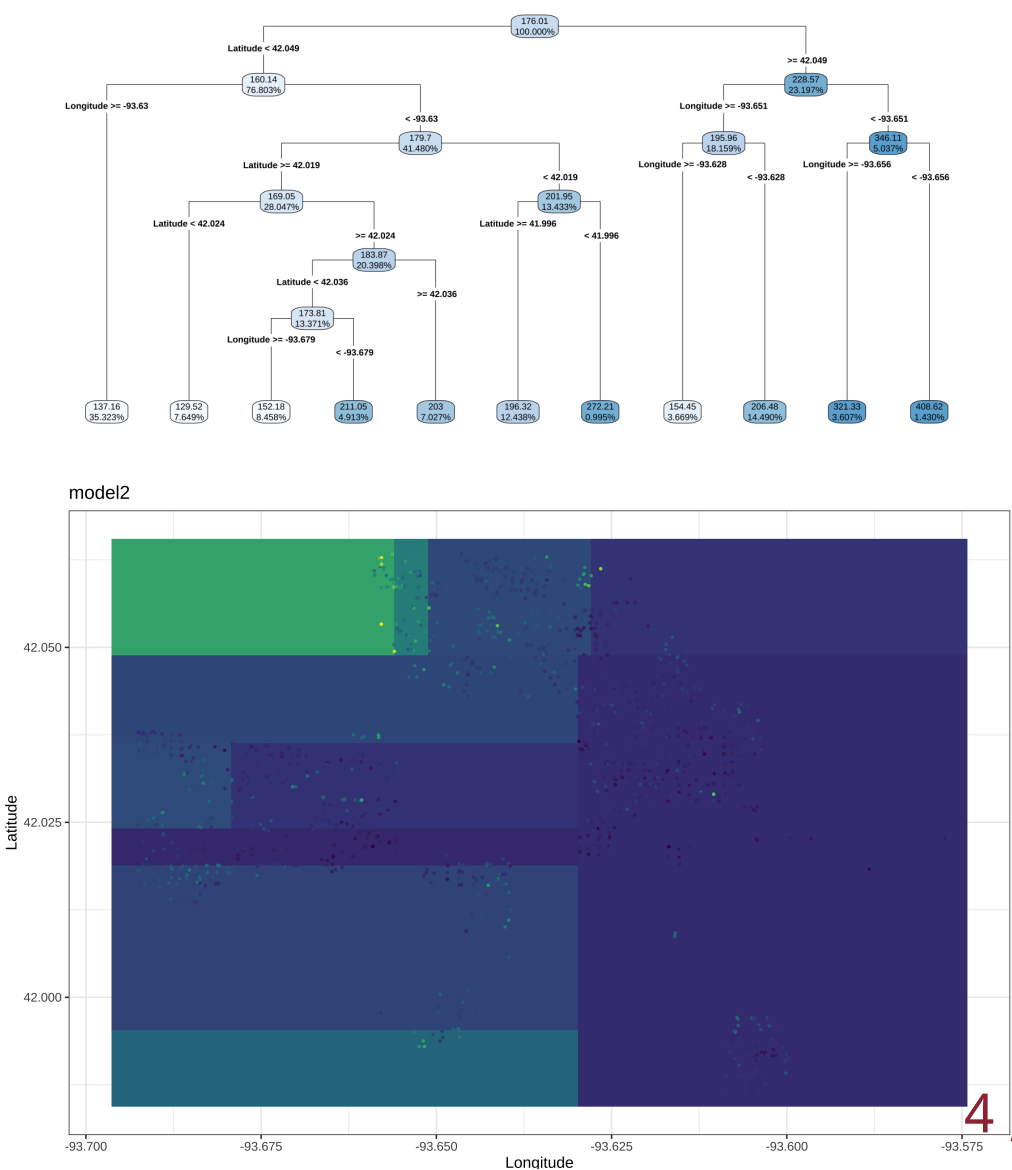
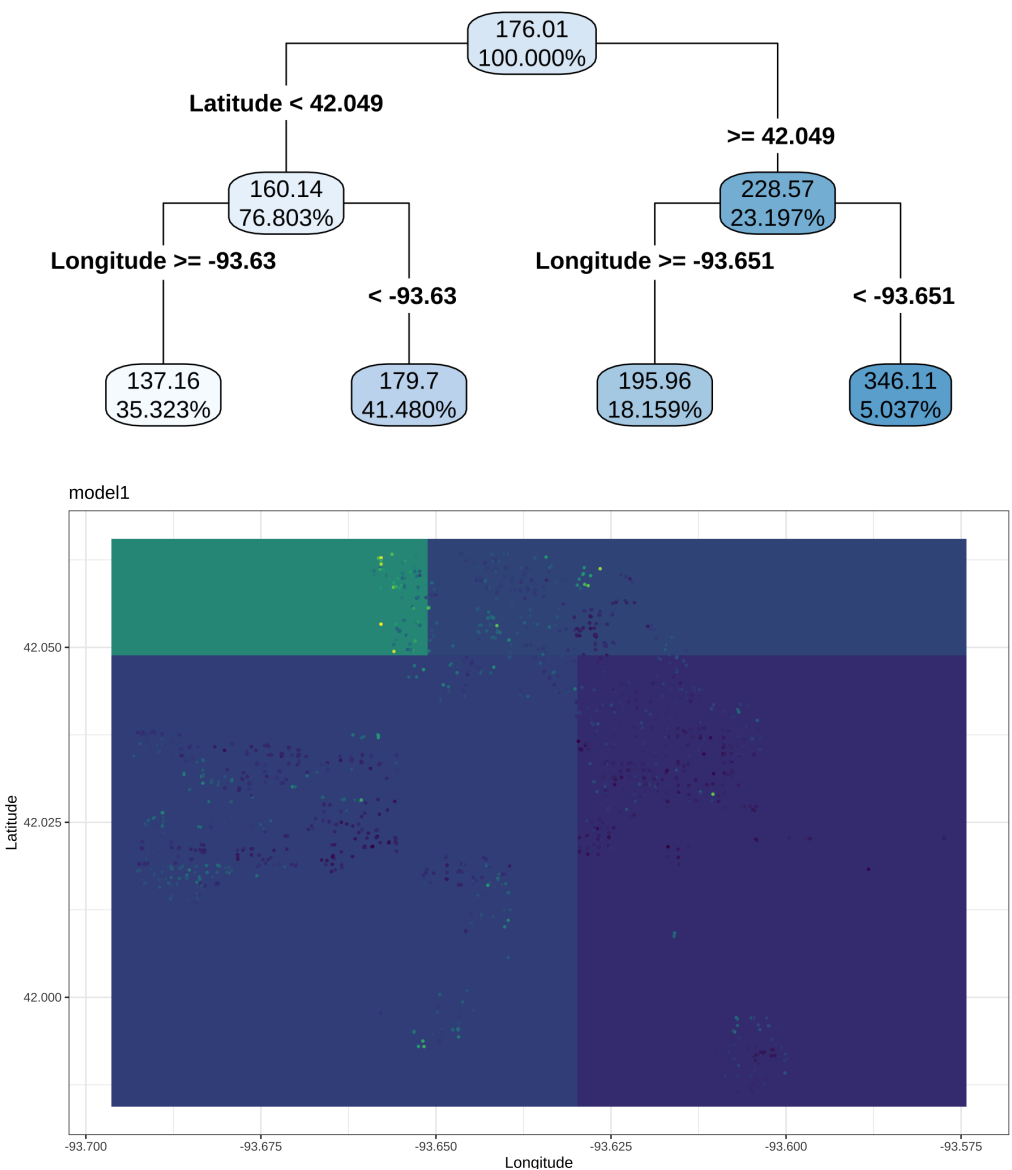
```
model2 <-
  decision_tree(mode = "regression", tree_depth = 30) %>%
  fit(Sale_Price ~ Latitude + Longitude, data = ames_train)
```

```
model3 <-
  decision_tree(mode = "regression", cost_complexity = 1e-6, min_n = 2) %>%
  fit(Sale_Price ~ Latitude + Longitude, data = ames_train)
```

This week

- *Today*: how modeling decisions affect performance; why validate?
- *Wednesday*: **how** to validate?
- *Friday*: cross-validation lab

Decision Trees



Q&A

| How do we *train* a decision tree?

- The model: "choose your own adventure": at each step, check one simple condition about one variable (e.g., `Latitude < 42.05`)
- Goal: find the best tree (for regression: minimize MSE)
- Approach: greedy algorithm: try all possible splits, keep the best one, repeat.

| I missed one of the check-in quizzes (or even weekly quizzes)!

Email me.

| Is midterm project individual?

At this point, yes.

Objectives

- Identify modeling decisions that affect the performance of decision tree and linear regression models, including:
 - Choice of model type
 - Pre-processing steps
 - Hyperparameter settings
- Explain the importance of *validation* for assessing and comparing models.

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- Shifting and scaling features

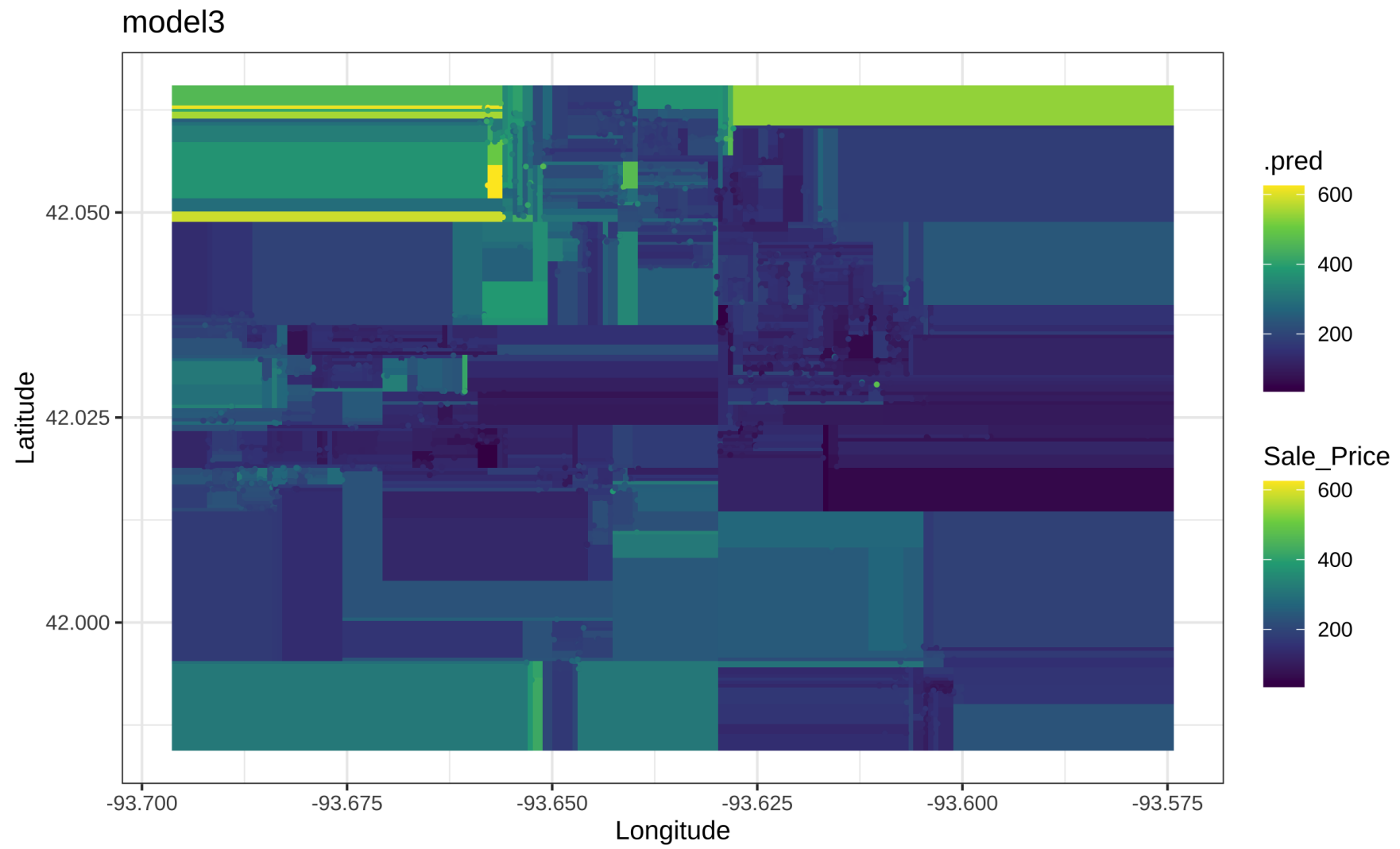
What are some decisions that we've made when making models so far?

- Which model to use
- Shifting and scaling features
- Hyperparameters: Tree depth, number of observations per leaf, ...

Hyperparameters for Decision Trees

```
model1 <-  
  decision_tree(mode = "regression", tree_depth = 2) %>%  
  fit(Sale_Price ~ Latitude + Longitude, data = ames_train)  
model2 <-  
  decision_tree(mode = "regression", tree_depth = 30) %>%  
  fit(Sale_Price ~ Latitude + Longitude, data = ames_train)  
model3 <-  
  decision_tree(mode = "regression", cost_complexity = 1e-6, min_n = 2) %>%  
  fit(Sale_Price ~ Latitude + Longitude, data = ames_train)
```

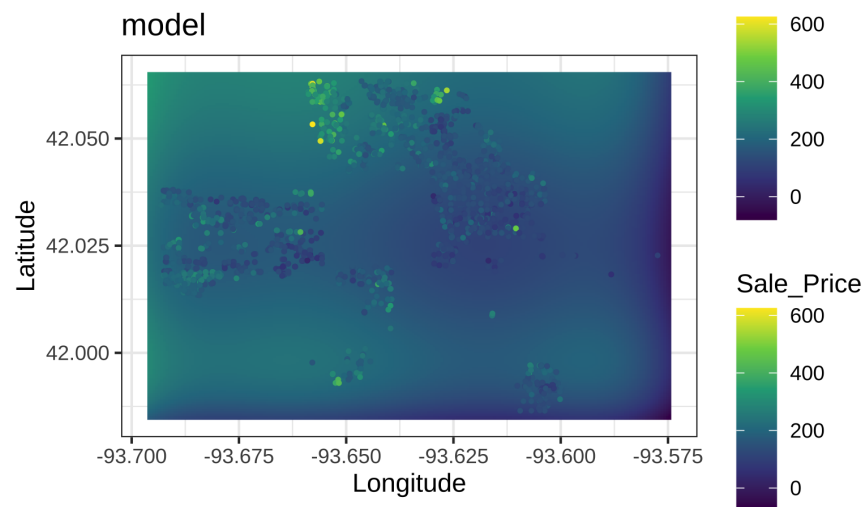
- Tree depth: how many levels of decisions
- Leaf size: how many observations need to be in each leaf node
- Complexity penalty: how much improvement for a split to be "worth it"



Hyperparameters for Linear Regression

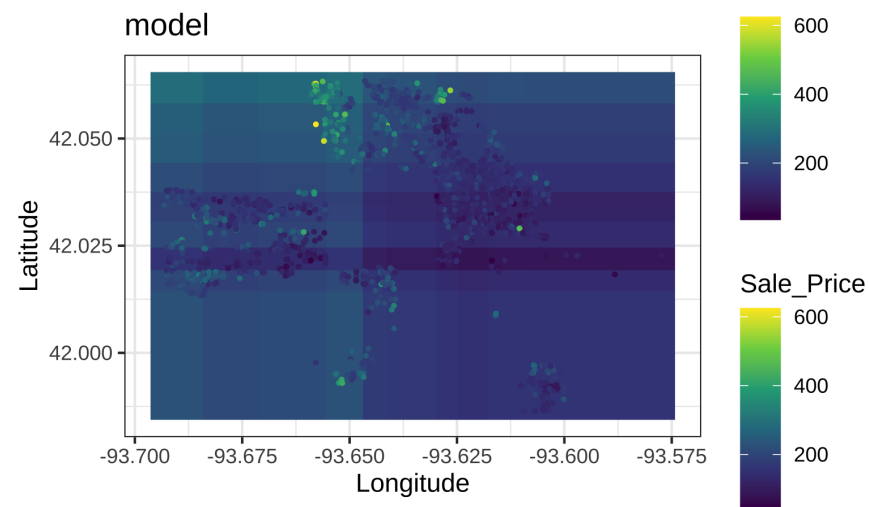
```
recipe <-  
  recipe(Sale_Price ~ Latitude + Longitude, data = ames_train) %>%  
  step_poly(Latitude, Longitude, degree = 5)
```

```
model <- workflow() %>% add_recipe(recipe) %>% add_model(linear_reg()) %>%  
  fit(ames_train)  
show_latlong_model(ames_train, model)
```



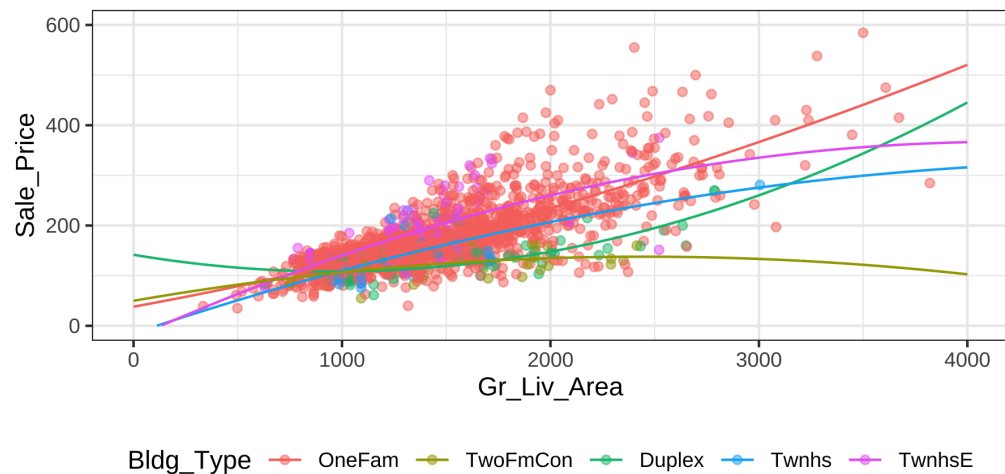
```
recipe <-  
  recipe(Sale_Price ~ Latitude + Longitude, data = ames_train) %>%  
  step_discretize(Latitude, Longitude, num_breaks = 5)
```

```
model <- workflow() %>% add_recipe(recipe) %>% add_model(linear_reg()) %>%  
  fit(ames_train)  
show_latlong_model(ames_train, model)
```



Polynomial and interaction features

```
recipe <-  
  recipe(Sale_Price ~ Gr_Liv_Area + Bldg_Type,  
    step_dummy(Bldg_Type) %>%  
    step_interact(  
      ~ Gr_Liv_Area:starts_with("Bldg_Type")) %>%  
    step_poly(  
      starts_with("Gr_Liv_Area"), degree = 2)
```



```
recipe <-  
  recipe(Sale_Price ~ Gr_Liv_Area + Bldg_Type,  
    step_dummy(Bldg_Type) %>%  
    step_interact(  
      ~ Gr_Liv_Area:starts_with("Bldg_Type")) %>%  
    step_poly(  
      starts_with("Gr_Liv_Area"), degree = 5)
```

